LEGACYLENS: An AI-Assisted Framework for Modernizing Legacy .NET Systems in Healthcare and Financial Services

Abstract

Legacy software systems continue to operate at the core of many healthcare, financial, insurance, and government organizations throughout the United States. While these systems often support mission-critical operations, they also create significant challenges related to cybersecurity, maintainability, technical debt, regulatory compliance, and workforce sustainability.

LegacyLens is an open-source AI-assisted modernization framework designed to help organizations analyze, understand, and transform legacy .NET applications into secure, maintainable, and cloud-ready systems.

Background

Many organizations remain dependent on software platforms built using aging technologies such as .NET Framework, ASP.NET MVC, Windows Communication Foundation (WCF), and other legacy architectures.

These systems often contain decades of undocumented business logic, outdated dependencies, security vulnerabilities, and operational constraints that make modernization costly and risky.

Although several commercial modernization tools exist, many are proprietary, vendor-locked, cloud-dependent, or focused on limited modernization scenarios.

This creates a gap between modernization needs and available modernization capabilities.

The LegacyLens Approach

LegacyLens combines deterministic software analysis with AI-assisted reasoning.

The framework is built around three foundational concepts:

1. Compiler-Grounded Analysis

Rather than relying exclusively on large language models, LegacyLens uses Microsoft's Roslyn compiler infrastructure to generate semantically accurate representations of software systems.

This allows the platform to understand types, dependencies, control flow, and architectural relationships before any AI-driven analysis occurs.

1. Business Rule Preservation

One of the most significant risks during modernization projects is the loss of institutional knowledge embedded in legacy code.

LegacyLens introduces a dedicated business-rule extraction process that identifies domain rules, workflows, constraints, and integration contracts before transformation activities begin.

1. Compliance-Aware Modernization

The framework incorporates modernization patterns aligned with common cybersecurity and compliance requirements, including:

• HIPAA • PCI DSS • OWASP Secure Coding Practices • Software Bill of Materials (SBOM) requirements • Secure cloud architecture principles

Expected Benefits

The project is expected to generate benefits across several dimensions.

Cybersecurity

Modernization can reduce vulnerability exposure by replacing obsolete technologies and introducing secure architectural patterns.

Operational Efficiency

Organizations may reduce technical debt, simplify maintenance efforts, and accelerate development cycles.

Knowledge Preservation

Business logic extraction reduces the risk of losing critical institutional knowledge during migration efforts.

Workforce Multiplication

By codifying modernization expertise into reusable workflows, LegacyLens may help organizations modernize systems even when experienced modernization specialists are scarce.

Broader Impact

The platform is being developed as an open-source initiative to maximize transparency, reproducibility, and accessibility.

Rather than benefiting a single employer or organization, the project is intended to provide reusable modernization methodologies that can be adopted by healthcare organizations, financial institutions, software vendors, government agencies, and academic researchers.

Future Development

The current roadmap includes:

Phase 1: Compiler-based analysis and dependency mapping.

Phase 2: Business-rule extraction and AI-assisted documentation.

Phase 3: Automated modernization orchestration and compliance-aware transformations.

Phase 4: Pilot implementations, validation studies, and publication of modernization case studies.

Conclusion

LegacyLens seeks to address one of the most persistent challenges facing modern software organizations: the modernization of aging, mission-critical software systems.

By combining compiler-level software understanding with AI-assisted modernization techniques, the project aims to improve cybersecurity, reduce modernization risk, preserve institutional knowledge, and support the long-term resilience of critical digital infrastructure throughout the United States.